

Olympiad Test Paper — Science (Class 7) — Paper 4

Chapter 15: Light

Paper 4 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	15 — Light
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Distractors weaponise the deepest misconceptions of the chapter: convex/concave mirrors swapped with convex/concave lenses, a converging device called diverging, virtual images projected onto screens, plane-mirror flips along the wrong axis, image distance scaled wrongly, a concave mirror said to always magnify, a convex mirror said to magnify, a prism said to create or add colours, red bending more than violet, a single pure colour split into a spectrum, and 'image grows as you approach' for a flat mirror. Do not write on this paper — mark answers on your answer sheet.

1. A lit candle stands on a table in a room with the curtains drawn. You can see the candle from anywhere in the room, but its sharp shadow on the wall has crisp straight edges only directly behind it. The crisp straight edge of the shadow, more than the fact that you can see the flame, is the cleaner proof that light:

- (A) travels in straight lines from the flame, so the candle blocks exactly the rays heading toward the wall behind it
- (B) spreads outward equally through opaque objects
- (C) is bent gently around the edges of the candle
- (D) changes colour as it passes the candle body

2. A narrow sunbeam enters through a slit and strikes a polished tabletop, throwing a bright patch on the ceiling; the same beam striking a velvet cloth throws no bright patch. At the exact spot where each ray meets its surface, the rule that the reflected ray leaves at the same angle to the normal as it arrived:

- (A) holds for both surfaces equally, since every individual ray obeys it even when the velvet's roughness scatters the rays in many directions
- (B) holds only for the polished tabletop and fails on the velvet
- (C) holds only for the velvet because rough surfaces reflect better
- (D) fails on both because the angle of incidence is always twice the angle of reflection

3. On a frosted (rough) sheet of glass you cannot see your face, yet on a clean window pane you faintly can. The frosted surface fails to give an image because it produces:

- (A) diffuse reflection, scattering the parallel rays into many directions so no organised image survives
- (B) regular reflection, which never forms a face image
- (C) total absorption of every ray that lands on it
- (D) dispersion that breaks the light into seven colours

4. A ray hits a plane mirror so that the angle between the ray and the mirror's surface is 25 degrees. The angle between the incident ray and the reflected ray (the total angle separating them) is:

- (A) 50 degrees
- (B) 25 degrees
- (C) 65 degrees
- (D) 130 degrees

5. A small lamp is set close behind a coin, casting a shadow on a far wall; then the lamp is moved far back behind the coin while the coin and wall stay put. As the lamp recedes, the dark central shadow of the coin on the wall:

- (A) becomes smaller, because the rays grazing the coin's edge run more nearly parallel and converge less past it
- (B) becomes larger, because the source is now farther away
- (C) vanishes, because distant light cannot cast a shadow
- (D) splits into the seven colours of the lamp

6. Three opaque cards each have a pinhole; a single lamp shines toward your eye through them. You see the lamp through all three holes only when:

- (A) the three holes are exactly equal in size
- (B) the cards are curved into a gentle arc
- (C) the holes are coloured to match the lamp
- (D) the three holes, the lamp and your eye all lie on one straight line

7. You stand 80 cm in front of a plane mirror holding a card 20 cm in front of your chest (so the card is 60 cm from the mirror). The distance from the card to the card's own image in the mirror is:

- (A) 160 cm
- (B) 60 cm
- (C) 100 cm
- (D) 120 cm

8. A girl walks toward a plane mirror at 1.5 metres per second. The gap between the girl and her image shrinks at a rate of:

- (A) 1.5 metres per second (B) 0.75 metres per second
(C) the gap does not change (D) 3 metres per second

9. A 1.7 m tall man stands before a plane mirror and then steps back to triple his original distance from it. The height of his image is now:

- (A) one-third of 1.7 m, since he tripled his distance
(B) still 1.7 m, because a plane-mirror image is always the same size as the object regardless of distance
(C) three times 1.7 m, matching his new distance
(D) 5.1 m, equal to the object-plus-image distance

10. A boy stands 3 m from a plane mirror; his friend stands 5 m from the same mirror. The distance between the boy's image and the friend's image (both measured behind the mirror) is:

- (A) 8 m (B) 16 m
(C) 10 m (D) 2 m

11. Two students aim to see each other's faces using one small flat mirror on a desk between them, with neither sitting opposite the mirror. They succeed only when the mirror is tilted so the path obeys equal angles to the normal. This works because a plane mirror:

- (A) sends every ray straight back the way it came
(B) reflects each ray so its angle to the normal on leaving equals its angle on arriving, letting light bend the corner from one face to the other
(C) curves light toward the taller student
(D) splits the face's light into colours along the way

12. A wall clock is reflected in a plane mirror and the mirror image appears to read 'ten minutes past eight' (8:10). The real time the clock shows is:

- (A) 3:50 (B) 8:10
(C) 4:50 (D) 2:10

13. A digital display built only from straight segments shows the figure made of '1' then '0' then '1'. Held up to a plane mirror, this three-digit display would appear:

- (A) as 101 with the zero flipped to a curve (B) as 010 with the digits swapped
(C) upside down
(D) unchanged as 101, because each of these digits is symmetric about a vertical line and the sequence reads the same reversed

14. The capital word 'WOW' is written on a transparent sheet and viewed normally through a plane mirror facing you. It appears:

- (A) as MOM, with the W's flipped into M's (B) as a backwards, unreadable string
(C) as WOW, unchanged, because every letter is symmetric about a vertical axis and the word reads the same reversed

(D) upside down as WOW rotated

15. On the front of a fire engine, the word 'FIRE' is painted so that a driver ahead reads it correctly in the rear-view mirror. As actually painted on the engine, the lettering is:

- (A) written normally, because mirrors do not change words
- (B) reversed left-to-right (mirror-writing), so the mirror's lateral inversion turns it back into normal FIRE
- (C) written upside down so the mirror flips it upright
- (D) written in colour so the mirror can read it

16. Of the capital letters A, F, J, T, exactly which set would look UNCHANGED in a plane mirror because each is symmetric about a vertical line down its middle?

- (A) all four
- (B) F and J only
- (C) A and T only
- (D) none of the four

17. A teacher writes a message on a glass door so that people OUTSIDE read it correctly, but a student standing inside reads it through the glass as reversed. To a person standing inside and viewing that reversed message in a plane mirror, it would appear:

- (A) doubly reversed and even harder to read
- (B) upside down but readable
- (C) correct and readable, because the mirror's lateral inversion undoes the reversal the student saw
- (D) unchanged, still reversed

18. A clock with no numbers, only hands, shows its minute hand pointing straight up and its hour hand pointing straight right (toward 3). In a plane mirror the reflected clock's hour hand points toward:

- (A) the 3 position (still straight right)
- (B) straight down toward 6
- (C) the 9 position (straight left)
- (D) straight up toward 12

19. A magician marks the spot where a candle's reflection appears to float behind a plane mirror, then slides a sheet of white card to that exact spot to catch the picture. No picture forms. The deepest reason is that:

- (A) no light actually reaches that spot; the rays only appear to diverge from there, so the image is virtual
- (B) the image is real but lands slightly off the marked spot
- (C) the card is too thin to hold the bright image
- (D) the image is inverted and cancels itself on the card

20. A convex lens forms a sharp, small, upside-down picture of a distant tower on a screen. A magnifying glass of the same kind held just above a stamp shows the stamp large and upright but throws nothing on a screen. The screen-catchable image and the screen-free image are, respectively:

- (A) both real, since a convex lens always makes real images
- (B) real and inverted for the distant tower; virtual and erect for the close stamp

- (C) virtual for the tower and real for the stamp
- (D) both virtual, since a lens never makes a screen image

21. Which one of these images could you actually catch on a sheet of paper held in the right place?

- (A) the enlarged print seen through a magnifying glass held close
- (B) your face in a flat bathroom mirror
- (C) a distant tree seen in a car's convex side mirror
- (D) the picture of a faraway window formed by a convex lens

22. A distant lamp post is viewed in a concave mirror; then a fingertip is brought to within a few millimetres of the same mirror. The lamp-post image and the fingertip image are, in order:

- (A) both virtual and erect
- (B) real and inverted for the far lamp; virtual, erect and enlarged for the very close fingertip
- (C) both real and inverted
- (D) virtual diminished for the lamp; real inverted for the fingertip

23. A student insists 'a concave mirror always makes things look bigger.' The accurate verdict is:

- (A) false, because a distant object gives a small, inverted, real image in a concave mirror
- (B) true for every position of the object
- (C) true only for brightly lit objects
- (D) false, because a concave mirror always shrinks things

24. A solar cooker uses a large mirror to gather sunlight onto a small cooking pot and heat it strongly. The correct mirror, and why, is:

- (A) a convex mirror, because it spreads the rays over a wide area
- (B) a plane mirror, because it keeps the rays the same size
- (C) a concave mirror, because it converges the parallel sunrays to a hot focus
- (D) a concave lens, because it focuses light to a point

25. As an object is carried steadily from far away right up to the surface of a concave mirror, the image it forms changes in the order:

- (A) real and inverted when far, then becoming virtual, erect and enlarged when very close
- (B) virtual erect when far, then real inverted when close
- (C) staying real and inverted at every distance
- (D) staying virtual and diminished at every distance

26. A dentist needs a mirror that, held close to a back tooth, shows it enlarged and the right way up. The correct mirror and the image it gives there are:

- (A) a convex mirror giving a magnified erect image
- (B) a concave mirror giving a virtual, erect, magnified image
- (C) a plane mirror giving an enlarged image up close
- (D) a concave mirror giving a real inverted image

27. A shop owner claims a corner security mirror 'magnifies thieves so they look bigger and easier to spot.' For a true convex security mirror this claim is:

- (A) correct, that is exactly why convex mirrors are chosen
- (B) correct only under bright shop lighting
- (C) wrong, because a convex mirror always forms a diminished image, so people look smaller while a wide area stays in view
- (D) wrong, because it inverts the image instead

28. A driver wants a single side mirror that shows the widest possible stretch of road behind while keeping every vehicle right-side-up. The best choice is:

- (A) a concave mirror, which gives a wide view and magnifies
- (B) a plane mirror, which gives the widest view of all
- (C) a convex mirror, which gives both a wide field of view and an erect image
- (D) a concave lens fitted as a mirror

29. A jeweller holds a hand lens thicker in the middle than at its edges just above a gemstone and sees it enlarged and upright. The lens is a:

- (A) convex (converging) lens
- (B) concave (diverging) lens
- (C) concave mirror
- (D) convex mirror

30. A convex lens is held far from a window and projects a tiny, upside-down picture of the trees outside onto a wall. This projected picture is:

- (A) virtual and erect
- (B) real and inverted
- (C) virtual and inverted
- (D) real and erect

31. Sunlight passed through a convex lens can burn a dot in a leaf, but the same-sized concave lens cannot scorch anything. The concave lens fails because it:

- (A) reflects the sunlight back toward the Sun
- (B) absorbs all the sunlight before it can pass
- (C) diverges the rays, spreading them apart instead of bringing them to a hot point
- (D) splits the sunlight into a harmless rainbow band

32. Looking at small print through a concave (diverging) lens, the letters appear:

- (A) larger and upright
- (B) larger and upside down
- (C) the same size but inverted
- (D) smaller and upright

33. Comparing a plane mirror, a concave mirror and a convex mirror all used on a DISTANT object, which statement is correct?

- (A) all three form real images of a distant object
- (B) only the convex mirror forms a real image
- (C) only the concave mirror can form a real image; the plane and convex mirrors always form virtual images
- (D) none of the three can ever form a real image

34. A polished steel spoon shows your face upside-down on its inner (bowl) side and right-way-up but smaller on its outer (back) side. The inner and outer surfaces act, respectively, as a:

- (A) convex mirror and a concave mirror (B) convex lens and a concave lens
- (C) plane mirror on both sides
- (D) concave mirror (real inverted image close to far) and a convex mirror (erect diminished image)

35. A make-up artist wants a mirror that gives an enlarged, upright view of the face when held close, while a delivery van wants one that shows a wide stretch of road behind. The pair of mirrors needed are, in order:

- (A) a convex mirror for the face and a concave mirror for the road
- (B) a plane mirror for both
- (C) a concave mirror for the close enlarged face and a convex mirror for the wide rear view
- (D) a concave lens for the face and a convex lens for the road

36. A lens that is thinner in the middle than at its edges is held over a printed line. The image and the lens type are:

- (A) an enlarged, erect image from a convex lens
- (B) a real, inverted image from a concave mirror
- (C) a diminished, erect image from a concave (diverging) lens
- (D) an enlarged, erect image from a concave lens

37. A torch maker places a reflector behind the bulb so the torch throws a strong, near-parallel beam down a path. The correct reflector and reason are:

- (A) a convex mirror, because it spreads the beam wide for safety
- (B) a plane mirror, because it keeps the beam the same width
- (C) a concave mirror, because with the bulb near its focus the reflected rays leave nearly parallel
- (D) a convex lens, because it converges the bulb's light to a point

38. Which single MIRROR shares with a plane mirror the property of always giving an ERECT image, yet unlike the plane mirror also makes the image SMALLER than the object whatever its position?

- (A) a concave mirror used on a near object (B) a convex mirror
- (C) a concave mirror used on a distant object (D) a concave lens

39. A child sorts four devices by whether they CONVERGE or DIVERGE light: a concave mirror, a convex mirror, a convex lens, a concave lens. The two that CONVERGE (bring rays together) are:

- (A) the convex mirror and the concave lens
- (B) the concave mirror and the concave lens
- (C) the concave mirror and the convex lens
- (D) the convex mirror and the convex lens

40. Four image descriptions are offered: (i) virtual, erect, same size; (ii) virtual, erect, diminished; (iii) real, inverted; (iv) real, erect and enlarged. Which one CANNOT be produced by any single mirror in this chapter?

- (A) (i) virtual, erect, same size
- (B) (ii) virtual, erect, diminished
- (C) (iii) real, inverted
- (D) (iv) real, erect and enlarged

41. A device gives an enlarged, upright image of close print AND can throw a small, upside-down picture of a far window onto a wall, BUT it is a lens, not a mirror. The device is a:

- (A) concave lens
- (B) concave mirror
- (C) convex lens
- (D) convex mirror

42. A teacher pairs four devices with the shape that describes them. Which pairing is entirely correct?

- (A) convex lens thinner in middle; concave lens thicker in middle; concave mirror bulging out; convex mirror curved in
- (B) convex lens curved inward; concave mirror thicker in middle; convex mirror thinner in middle; concave lens flat
- (C) all four are simply flat plates of glass or metal
- (D) convex lens thicker in middle; concave lens thinner in middle; concave mirror curved inward; convex mirror bulging outward

43. Which row correctly states the image each device forms?

- (A) plane mirror gives virtual same-size; convex mirror gives virtual diminished; concave lens gives virtual diminished
- (B) plane mirror gives real same-size; convex mirror gives virtual magnified; concave lens gives real
- (C) concave mirror on a distant object gives virtual erect; convex mirror gives real; plane mirror gives inverted
- (D) convex lens always gives a virtual image; concave mirror never gives a real image; convex mirror magnifies

44. A camera lens that brings light from a far scene to a sharp focus on the film, a magnifying glass, and a torch's concave reflector are grouped by what they do to parallel rays. The odd one out is:

- (A) the magnifying glass, because it only diverges light
- (B) none of them is odd; all three converge parallel rays

- (C) the camera lens, because it spreads light out
- (D) the torch reflector, because it scatters light randomly

45. White sunlight enters a glass prism and emerges as a band of colours, while the same sunlight through a flat-faced pane of window glass shows no such band. The prism disperses but the pane does not because the prism's two refracting faces are:

- (A) coloured, which adds the colours to the white light
- (B) made of a special rainbow glass
- (C) curved like a lens, which creates the colours
- (D) slanted (not parallel), so the colours, once separated, stay spread apart instead of recombining

46. Reading the prism's spectrum from the colour that bends LEAST to the colour that bends MOST, the order is:

- (A) red, orange, yellow, green, blue, indigo, violet
- (B) violet, indigo, blue, green, yellow, orange, red
- (C) red, yellow, blue only
- (D) green, red, violet, blue, orange, yellow, indigo

47. A student writes that 'violet bends the least and red bends the most through a prism.' The correct version is that:

- (A) red bends the least and violet bends the most
- (B) both bend exactly equally
- (C) red bends most and orange bends least
- (D) the bending order depends on how bright the light is

48. Pure orange light, a single colour, is passed through a glass prism. What emerges is:

- (A) the full seven-colour spectrum
- (B) only orange light, bent a little but not split into other colours
- (C) white light
- (D) no light at all

49. A glass prism splits white light into seven colours; a disc painted with those seven colours and spun very fast looks whitish. Taken together these two observations show that:

- (A) the prism creates the seven colours out of plain white
- (B) white light is made of the seven colours, which can be split apart by the prism and recombined by the spinning disc
- (C) white light is a single pure colour
- (D) the spinning disc destroys the colours

50. After a rain shower with the Sun low behind you, a rainbow arches across the sky. Each raindrop in the arc behaves like a tiny:

- (A) mirror, reflecting white light back unchanged
- (B) magnifying lens that enlarges the Sun
- (C) filter that lets only one colour through
- (D) prism, splitting the sunlight into its colours

51. A teacher claims, 'A rainbow proves the Sun makes new colours after rain.' The correct response is that:

- (A) no new colours are made; the raindrops only separate colours already present in the white sunlight
- (B) yes, raindrops manufacture the seven fresh colours
- (C) yes, but only red and blue are newly made
- (D) the rainbow's colours come from the blue sky, not the Sun

52. A flat pane of glass with two parallel faces is held in a sunbeam and produces no band of colours, even though it bends the light slightly as it passes. The best reason no spectrum appears is that:

- (A) flat glass cannot bend light at all
- (B) the parallel faces let any colours that begin to separate recombine on the way out, so they emerge overlapping as white
- (C) flat glass absorbs every colour except white
- (D) flat glass reflects all the light back, leaving none to disperse

53. Three demonstrations are shown: (i) a still lake reflecting a mountain, (ii) sunlight through a prism making a colour band, (iii) a hand lens projecting an upside-down picture of a window onto a wall. They illustrate, in order:

- (A) dispersion, reflection, and a virtual image from a concave lens
- (B) refraction, dispersion, and reflection
- (C) diffuse reflection, lateral inversion, and magnification
- (D) reflection, dispersion, and a real image from a convex lens

54. A spinning Newton's disc painted with the seven spectrum colours looks whitish, while the same disc held still shows the separate colours. Stopping the disc changes the appearance because:

- (A) the colours disappear when the disc is moving
- (B) the eye no longer blends the rapidly arriving colours, so the separate paints become visible again
- (C) the disc creates white paint when it spins
- (D) the still disc reflects only white light

55. A laser beam crossing a dusty room is a perfectly straight bright line. A thin flat glass slab slipped into the beam shifts the line sideways but leaves it straight on the far side. The unbroken straightness across the room is evidence that, within one uniform medium, light:

- (A) travels in straight lines
- (B) curves gently downward over distance
- (C) spreads into a widening cone
- (D) splits into its seven colours by itself

56. A teacher lists three uses and asks which mirror type ALL THREE share: a shaving mirror, a solar-cooker reflector, and a car-headlight reflector. The shared mirror is:

- (A) the concave mirror
- (B) the convex mirror
- (C) the plane mirror
- (D) a mix: shaving uses convex, cooker uses plane, headlight uses concave

57. Which complete chain of reasoning is entirely correct for a convex MIRROR?

- (A) it converges reflected rays, so it forms a real inverted image used in torches
- (B) it always magnifies nearby objects, which is why it suits security mirrors
- (C) it diverges reflected rays, so the image is always virtual, erect and diminished, giving a wide field of view useful as a rear-view mirror
- (D) it can throw a real image onto a screen when the object is far away

58. A complete statement about converging and diverging behaviour is needed. Which one is correct throughout?

- (A) a concave mirror and a concave lens both converge, while the convex ones both diverge
- (B) a concave mirror and a convex lens both converge light, while a convex mirror and a concave lens both diverge it
- (C) a convex mirror and a convex lens both converge, while the concave ones both diverge
- (D) all four devices converge light to a point

59. A final synthesis pairs each device with the kind of image it can form. Which row is correct in every part?

- (A) plane mirror gives real inverted; convex mirror gives real magnified; concave lens gives real
- (B) concave mirror (distant object) gives real inverted; convex mirror gives virtual diminished; convex lens (distant object) gives real inverted
- (C) concave mirror always gives virtual erect; convex lens always gives virtual; concave lens gives real
- (D) convex mirror gives real on a screen; plane mirror gives inverted; concave mirror never gives a real image

60. A wide lamp (not a point source) shines past a small ball onto a wall, casting a shadow with a fully dark core surrounded by a fuzzy grey rim. The fuzzy grey rim exists because, at those edge points on the wall:

- (A) the ball bends light around itself to soften the edge
- (B) the wall colours the light grey at the boundary
- (C) the light slows down and spreads as it passes the ball
- (D) part of the wide lamp is still visible past the ball, so only some of its rays are blocked, partially lighting the region

Solutions — Science (Class 7), Chapter 15: Light — Paper 4

Paper 4 — (progressively harder)

Marking: +4 correct, −1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	11	B	21	D	31	C	41	C	51	A
2	A	12	A	22	B	32	D	42	D	52	B
3	A	13	D	23	A	33	C	43	A	53	D
4	D	14	C	24	C	34	D	44	B	54	B
5	A	15	B	25	A	35	C	45	D	55	A
6	D	16	C	26	B	36	C	46	A	56	A
7	D	17	C	27	C	37	C	47	A	57	C
8	D	18	C	28	C	38	B	48	B	58	B
9	B	19	A	29	A	39	C	49	B	59	B
10	D	20	B	30	B	40	D	50	D	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (A) Seeing the flame only proves light reaches your eye; it is the geometric sharpness of the shadow's edge that shows the rays move in straight paths and are cleanly cut off by the body. Light cannot pass through an opaque candle, it is not bent around its edges to soften them, and no colour change occurs.

2. (A) The law of reflection is obeyed point-by-point on any surface; velvet looks dull because its tiny facets face random directions, scattering rays, but each ray still reflects symmetrically. So it cannot be true for one surface only, and the angle of incidence equals the angle of reflection, never double it.

3. (A) A rough surface sends neighbouring rays off at differing angles, destroying the orderly bundle needed for an image; this scattering is diffuse reflection. Regular reflection is exactly what forms images, the glass is not absorbing all the light, and no colour-splitting is happening.

- 4. (D)** The angle from the normal is 90 minus 25, that is 65 degrees, for both the incident and reflected rays; the two rays together span 65 plus 65, which is 130 degrees. Taking 50 doubles the surface angle wrongly, 25 is only the grazing angle, and 65 is just one of the two equal halves.
- 5. (A)** A far source sends near-parallel rays, so the cone of shadow narrows and the shadow shrinks toward the coin's true size; a near source spreads the shadow wide. A distant source still casts a shadow, and shadows are colourless, so they do not split into a spectrum.
- 6. (D)** Because light travels straight, a ray can thread all three holes to your eye only if they fall on a common straight line with the source. Equal hole sizes, a curved arrangement, or matching colours do not by themselves let the straight ray pass.
- 7. (D)** The card's image sits as far behind the mirror as the card is in front, namely 60 cm, so card-to-image is 60 plus 60, which is 120 cm. The 160 figure uses your 80 cm distance, 60 forgets the equal image distance, and 100 mixes the two object distances.
- 8. (D)** Both she and her image approach the mirror at 1.5 metres per second, so they close on each other at the sum, 3 metres per second. Quoting 1.5 counts only her motion, 0.75 halves it, and the gap certainly changes since both are moving.
- 9. (B)** Distance from a plane mirror never changes image size; it only changes how big the image looks to the eye, not its true size, which stays equal to the object. So it neither shrinks to a third, nor grows threefold, nor reaches any summed distance value.
- 10. (D)** The two images lie 3 m and 5 m behind the mirror, so they are 5 minus 3, that is 2 m, apart. Adding gives 8, doubling each and adding gives 16, and 10 doubles the friend's distance alone.
- 11. (B)** A flat mirror obeys equal angles point by point, so a correctly tilted mirror routes light from one student's face to the other's eye. It does not retro-reflect straight back, it does not favour a student by height, and a plane mirror does not disperse light.
- 12. (A)** A mirror clock is read by taking 12:00 minus the apparent time: 12:00 minus 8:10 gives 3:50. Reading it directly as 8:10 ignores the reversal, and 4:50 and 2:10 come from subtracting from the wrong reference.
- 13. (D)** The digits 1 and 0 are each left-right symmetric and the string 101 reads the same backwards, so the mirror leaves it looking identical. A mirror flips left-right not the digit shapes' symmetry, does not swap 1 and 0, and does not turn things upside down.
- 14. (C)** W and O are each vertically symmetric and WOW is a palindrome, so the left-right mirror flip leaves it identical. A vertical mirror does not invert a W into an M (that is a top-bottom flip), does not scramble the symmetric letters, and does not rotate the word.

15. (B) Because the rear-view mirror laterally inverts, the engine carries the pre-inverted (backwards) form so the reflection comes out readable. Normal lettering would appear backwards in the mirror, an upside-down trick addresses the wrong axis, and colour has nothing to do with readability.

16. (C) A and T are symmetric about a vertical centre line, so they survive the left-right flip; F and J are not vertically symmetric and appear reversed. Hence not all four, not the asymmetric F and J, and not none.

17. (C) The student inside already sees a left-right reversed message; a plane mirror reverses left-right once more, restoring the original readable text. It does not reverse it twice in the same sense, it does not flip top-bottom, and the mirror does change it rather than leaving it reversed.

18. (C) A vertical mirror swaps left and right but not up and down, so a hand pointing right (3) reflects to point left (9), while a vertical hand stays vertical. It does not stay at 3, does not flip to 6 (that needs a horizontal mirror), and the up-pointing minute hand, not the hour hand, stays at 12.

19. (A) A plane-mirror image is virtual: reflected rays merely seem to come from behind the mirror, where no real light converges, so nothing can be captured. The image is not real, card thickness is irrelevant, and a plane mirror image is erect, not self-cancelling.

20. (B) A convex lens gives a real inverted image of a far object (catchable) and a virtual erect enlarged image of a very near one (not catchable). So it is not real in both cases, the pairing is not swapped, and a convex lens certainly can make a real image.

21. (D) Only a real image, such as the convex lens's image of a far object, can land on a screen. The magnifier's close-up image, the plane-mirror face, and the convex-mirror view are all virtual and so cannot be projected.

22. (B) A concave mirror inverts and reduces a far object into a real image, but a very near object gives a virtual, erect, magnified image. So the two are not both of one kind, and the pairing must not be reversed.

23. (A) A concave mirror enlarges only a close object; a far object gives a diminished, inverted real image, so the 'always bigger' claim fails. It is neither true for all positions nor brightness-dependent, and it does not always shrink either.

24. (C) Concentrating sunlight to a hot spot needs converging, which a concave mirror does by bringing parallel rays to a focus. A convex mirror diverges, a plane mirror neither focuses nor concentrates, and a concave lens diverges rather than focuses (and is a lens, not a mirror).

- 25. (A)** A concave mirror gives a real, inverted image for distant objects and switches to a virtual, erect, enlarged image once the object is very near. The sequence is therefore not reversed, and it does not stay one single type throughout.
- 26. (B)** Only a concave mirror, with the tooth very close to it, produces an enlarged, erect, virtual image, which is what a dentist wants. A convex mirror only diminishes, a plane mirror keeps true size, and a concave mirror inverts only for far objects.
- 27. (C)** A convex mirror always gives a smaller, erect, virtual image; its value is the wide field of view, not magnification. So the claim is not correct, not lighting-dependent, and the image is erect, not inverted.
- 28. (C)** A convex mirror diverges reflected rays to take in a wide scene and always keeps the image erect. A concave mirror narrows the view and can invert, a plane mirror's view is not wider than a convex one of the same size, and a lens is not a mirror.
- 29. (A)** A lens thicker in the middle is convex; held close, it gives an enlarged, erect, virtual image, like a magnifying glass. A concave lens always diminishes, and the two mirror options are mirrors, not lenses.
- 30. (B)** A distant object through a convex lens forms a real, inverted image that can fall on a wall. A virtual image cannot be projected, and the projected image of a far object through a single convex lens is inverted, not erect.
- 31. (C)** A concave lens spreads parallel rays outward, so they never concentrate to a burning focus. It transmits rather than reflects the light, does not absorb it all, and a flat-faced lens does not disperse sunlight into a rainbow.
- 32. (D)** A concave lens always forms a virtual, erect, diminished image, so the print looks smaller and upright. It does not enlarge, does not invert close print, and changes the size rather than leaving it unchanged.
- 33. (C)** Of the three mirrors only a concave one can converge reflected light to a real image; plane and convex mirrors give virtual images for any object. So not all three, not the convex, and the claim that none can is also false.
- 34. (D)** The hollow inner bowl curves inward like a concave mirror and can invert; the bulging back curves outward like a convex mirror and always gives an erect, diminished image. The pairing is not reversed, neither surface is a lens, and a curved spoon is not flat.
- 35. (C)** Close-up magnification needs a concave mirror; a wide field of view behind a vehicle needs a convex mirror. The two roles are not swapped, a plane mirror neither magnifies nor widens, and lenses are not mirrors.

36. (C) A lens thinner in the middle is concave and always gives a diminished, erect, virtual image. A convex lens (thicker middle) would enlarge, a mirror is not a lens, and a concave lens never enlarges.

37. (C) A concave mirror with the source at its focus sends out a tight, near-parallel beam, which is what a torch needs. A convex mirror would spread the light, a plane mirror would not concentrate it, and a lens is not a reflector.

38. (B) A convex mirror always gives an erect but diminished image, matching the plane mirror's erectness while shrinking the object at every position. A near-object concave mirror enlarges, a distant-object concave mirror inverts (not erect), and a concave lens is a lens, not a mirror as the question demands.

39. (C) A concave mirror and a convex lens both bring parallel rays toward a point, so they converge; the convex mirror and concave lens both spread rays, so they diverge. The other pairings each mix one converging with one diverging device.

40. (D) A plane mirror gives virtual same-size, a convex mirror gives virtual diminished, and a concave mirror gives real inverted, but no chapter mirror gives a real image that is also erect. So the impossible one is the real-and-erect-enlarged case.

41. (C) A convex lens magnifies near print (virtual erect) and projects a far object as a real inverted image, and it is a lens. A concave lens only diminishes, while the two mirror options are mirrors, not lenses.

42. (D) A convex lens is thick in the middle, a concave lens thin in the middle, a concave mirror dips inward, and a convex mirror bulges outward, so the first pairing is right throughout. The others swap lens thicknesses or mirror curvatures, or wrongly call curved devices flat.

43. (A) A plane mirror image is virtual and same-size, a convex mirror image is virtual and diminished, and a concave lens image is virtual and diminished, so that row is fully correct. The others wrongly call a plane image real, a convex image magnified, a distant concave image virtual, or a convex lens always-virtual.

44. (B) A focusing camera lens, a magnifying glass and a concave reflector are all converging devices, so none stands apart. The magnifier does not diverge, the camera lens does not spread light, and a concave reflector gathers rather than scatters.

45. (D) A prism's non-parallel faces keep the separated colours apart, while a slab's parallel faces let them rejoin so no band shows. The prism does not add colour, is ordinary glass, and is not curved like a lens; the colours come from the white light itself.

46. (A) Red bends least and violet bends most, so least-to-most runs red, orange, yellow, green, blue, indigo, violet. The reverse lists most-to-least, three colours is incomplete, and

the scrambled list is out of order.

47. (A) Through a prism violet is deviated most and red least, the opposite of the student's claim. The colours do not bend equally, the red-most ordering is wrong, and the bending order does not depend on brightness.

48. (B) A prism can only separate colours that are already present; a single colour comes out as that colour, merely deviated. It cannot create a spectrum from one colour, cannot turn one colour into white, and does not block the light.

49. (B) The prism separates and the disc recombines, proving white light is the seven colours mixed together. The prism does not create colour, white is not a single pure colour, and the disc blends rather than destroys the colours.

50. (D) Each droplet refracts and disperses sunlight like a small prism, producing the rainbow's colours. A mirror would not split colours, a lens would magnify not disperse, and a filter would remove colours rather than spread them.

51. (A) Rainbow colours are simply the components of white sunlight pulled apart by raindrops; nothing new is created. So the droplets neither manufacture colours nor make a subset of them, and the colours originate in the sunlight, not the sky.

52. (B) With parallel faces, the slight separation entering the slab is reversed leaving it, so the colours overlap back into white. The glass does bend the light, does not absorb colours, and does not reflect it all away.

53. (D) A still lake reflects, a prism disperses, and a convex lens projecting an inverted picture forms a real image, matching the order. The other rows mislabel one or more demonstrations or assign the wrong device.

54. (B) Fast spinning makes the colours reach the eye too quickly to be seen apart, so they blend to white; stationary, each colour is seen on its own. The colours do not vanish in motion, no white paint is created, and the still disc shows colours, not white.

55. (A) A straight beam in a single medium shows light moves in straight lines; the slab only shifts it, without bending the overall path. Light does not droop, does not fan into a cone in a uniform medium, and does not self-disperse.

56. (A) Shaving mirrors, solar cookers and headlight reflectors all use concave mirrors, for magnification or for converging and beaming light. They are not convex or plane, and they are not a mix of types.

57. (C) A convex mirror spreads reflected rays, always giving a virtual, erect, diminished image and a wide view, ideal for rear-view use. It does not converge or form real images, and it never magnifies.

58. (B) Converging devices are the concave mirror and convex lens; diverging ones are the convex mirror and concave lens, so the first statement is right. Grouping by name alone (both concave, both convex) is wrong, and not all four converge.

59. (B) A distant object gives a real inverted image in both a concave mirror and a convex lens, while a convex mirror always gives a virtual diminished image, so that row holds throughout. The others wrongly make plane images real, convex-lens images always virtual, or convex-mirror images real.

60. (D) A broad source means the ball blocks all rays only over a small core, while around it some parts of the lamp peek past, giving partial light, the grey rim. Light is not bent around the opaque ball, the wall does not tint it, and rays do not slow and spread in air to make the fuzziness.